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2. RELATIONS

Cartesian product (or) product set (or) cross product (x):

If A and B are two non-empty set we define the cartesian product A×B is the set of all ordered pairs (a,b) where $a \in A$, $b \in B$.

$\therefore A \times B = \{(a,b) \mid a \in A \text{ \& } b \in B\}$

Ex: $A = \{1,2,3\}$ $B = \{q, s\}$

$A \times B = \{(1,q), (2,q), (3,q), (1,s), (2,s), (3,s)\}$

Note: $A \times B \neq B \times A$

Let A and B be a non-empty set, a relation R from A to B is a subset of A×B. i.e $R \subset (A \times B)$ and $(a,b) \in R$ and we say that aRb.

If A is not related B by R $a \not R b$

If A and B are equal ie $R \subset A \times A$. aRa.

Eg: $A = \{1,2,3\}$ $B = \{q,s\}$

$R = \{(1,q), (2,s), (3,q)\}$

1. If A and B are set of a real no. then we define the relation R' from A→B aRb iff a=b. Let $A = \{1,2,3,4,5\}$

Define a relation R is less than A when A is related to B

⇒ $R = \{(1,2), (1,3), (1,4), (1,5), (2,3), (2,4), (2,5), (3,4), (3,5), (4,5)\}$

2. Let $A = \{a,b,c\}$ and $B = \{1,2,3\}$

(i) Is $R = \{(a,1), (b,2), (c,2)\}$ a relation from A to B — True

(ii) Is $Q = \{(1,a), (2,b)\}$ a relation from A to B — False.

(iii) Is $P = \{(a,a), (b,a), (b,a)\}$ a relation from A×A — True.

3. Let $A = \{1,2,3,4\}$ $R = \{(a,b) \mid a \text{ divides } b\}$

⇒ $R = \{(1,2), (1,3), (1,4), (2,4), (1,1), (2,2), (3,3), (4,4)\}$

Properties & Types of relation:

1. Reflexive:
   A relation R on set A is said to be reflexive if for all the elements of $a \in A$ where $(a,a) \in R$.
   Ex: $(a,b,c) \times (a,b,c)$
   $R = \{(a,a), (b,b), (c,c)\}$

1. $R = \phi$ No
2. $R = A \times A$ Yes
3. $R = \{(a,a), (b,b), (c,c)\}$ Yes

2. Irreflexive:
   A relation R on set A is said to be irreflexive if $\forall a \in A$, $(a,a) \notin R$.

3. Symmetric:
   A relation R on set A is said to be symmetric if $\forall (a,b) \in R$, $(a,b) \in R$ & $(b,a) \in R$.
   Ex: ① $R = \{(a,b), (b,a)\}$
   ② $R = \{(b,c), (c,b), (b,b), (c,c)\}$
   ③ $R = \{(a,a), (b,b), (c,c)\}$

1. $R = \phi$ No
2. $R = A \times A$ Yes
3. $R = \{(a,b), (b,c), (a,c)\}$ No
4. $R = \{(a,b), (b,a), (a,c)\}$ No

4. Assymmetric:
   A relation R on set A is said to assymmetric if $\forall (a,b) \in R$, $(a,b) \in R$ & $(b,a) \notin R$.

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1. $\{(a,b),(b,c),(c,a)\}$ Yes
2. $\{(a,b),(b,c),(c,b)\}$ yes
3. $\{(a,a),(b,b),(c,c)\}$ No
4. $A \times A$ No
5. $\{(a,b),(a,c),(b,c)\}$ Yes.

5. Anti-symmetric :

A relation R on set-A is said to be anti-symmetric if $\forall (a,b) \in A$ where as $(a,b) \in R$ & $(b,a) \in R$ if $a=b$.

1. $\{(a,b),(b,c),(a,c)\}$ True
2. $\{(a,b),(a,a),(b,b)\}$ True
3. $\{(a,a),(b,b),(c,c)\}$ True

6. Transitive :

A relation R on set A is said to be transitive if $\forall a,b \in A$, $(a,b) \in R$, $(b,c) \in R$ then $(a,c) \in R$.

1. $\{(a,b),(b,a),(a,a),(b,b)\}$
2. $\{(a,a),(b,b),(c,c)\}$
3. $\{(a,b)\}$
4. $\{(a,b),(a,c)\}$
5. $\{(a,b),(c,b)\}$
6. $\{(a,b),(b,c),(a,c),(a,a)\}$
7. $\{(c,a),(b,c)\}$

Yes.

1. Let $A=\{1,2,3\}$, Determine the nature on the following relation on A

$R_1 = \{(1,2),(2,1),(1,3),(3,1)\}$ → Symmetric, irreflexive

$R_2 = \{(1,1),(2,2),(3,3),(2,3)\}$ → Reflexive, transitive.

$R_3 = \{(1,1),(2,2),(3,3)\}$ → Reflexive, symmetric, reflexive.

$R_4 = \{(1,1),(2,2),(3,3),(2,3),(3,2)\}$ → Reflexive, transitive.

$R_5 = \{(1,1),(2,3),(3,3)\}$ → Transitive

$R_6 = \{(2,3),(3,4),(2,4)\}$ Transitive.

$R_7 = \{(1,3),(3,2)\}$ Irreflexive.

Matrix of a relation [MR] or [MR] or zero:

Consider a set $A = [A_1, A_2, A_3, \ldots, A_m]$ & $B = [B_1, B_2, B_3, \ldots, B_n]$ of order m×n respectively and R is a relation from A to B then R is represented by the matrix m×n i.e $M_R = [m_{ij}]$ where

$$m_{ij} = \begin{cases} 1, & \text{iff } (a_i, b_j) \in R \\ 0, & \text{iff } (a_i, b_j) \notin R. \end{cases}$$

So the matrix framed by m_{ij} is called the relation matrix for R and it is denoted by [MR] or [MR]

Since the [MR] contains only 0's and 1's it is also called as zero-one matrix for a relation.

Ex: $A = \{1,2,3\}$ & $B = \{a,b\}$

$R = \{(1,a),(2,b),(3,a)\}$

$$M_R = \begin{array}{c} \\ 1 \\ 2 \\ 3 \end{array} \begin{array}{c} \begin{array}{cc} a & b \end{array} \\ \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \end{array}$$

1. Consider a matrix $M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$ where $A = \{a_1, a_2, a_3\}$ and $B = \{b_1, b_2, b_3, b_4\}$. Write the relation.

Solution: $A = \{a_1, a_2, a_3\}$ $B = \{b_1, b_2, b_3, b_4\}$

$$M_R = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

$R = \{(a_1,b_1), (a_1,b_4), (a_2,b_2), (a_2,b_3), (a_3,b_1), (a_3,b_3)\}$

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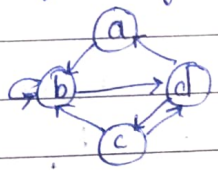
Diagraph:

Let R be a relation on a finite set A.
Let R can be represented by pictorically.
Draw a small circle or a bullet for each element of A and label the circle with the corresponding elements of A. This circles or bullets are called vertices or nodes. Draw an arrow called the edge from a vertex x to vertex y iff $(x,y) \in R$ so the resulting pictorical representation of R is called a directed graph or diagraph of a relation.

The no. of edges (arrows) terminating at a vertex is called the indegree of the vertex, no. of edges leaving a vertex is called the outdegree of the vertex.

1. Consider $A = \{a, b, c, d\}$ and $R = \{(a,b), (b,b), (b,d), (c,b), (c,d), (d,a), (d,c)\}$ define on A. Write the diagraph of the relation and the no. of indegree and outdegree of the relations

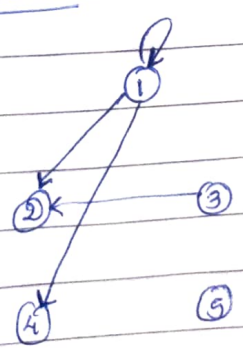
Solution:



	a	b	c	d
Indegree	1	3	1	2
Outdegree	1	2	2	1

2. $A = \{1, 2, 3, 4, 5\}$ and $R = \{(1,1), (1,2), (1,4), (3,2)\}$

Solution:

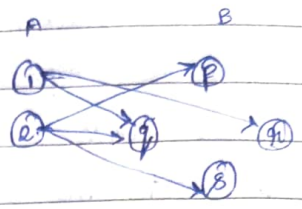


	1	2	3	4	5
Indegree	1	2	0	1	0
Outdegree	3	0	1	0	0

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3. $A = \{1, 2\}$, $B = \{p, q, r, s\}$ and $R = \{(1,q), (1,r), (2,p), (2,q), (2,s)\}$

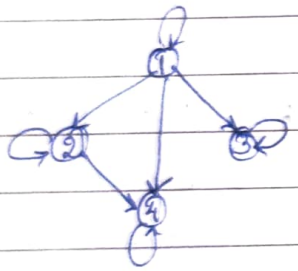
$$M_R = \begin{matrix} \\ 1 \\ 2 \end{matrix} \begin{matrix} p & q & r & s \end{matrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \end{bmatrix}$$



4. Let $A = \{1, 2, 3, 4\}$ and R be the relation on A defined by x related to y iff x divides y.
   (i) Write down the R as a set of ordered pairs.
   (ii) Draw the diagraph of the relation and determine the indegree and outdegree of the relation in the diagraph.

Solution: $A = \{1, 2, 3, 4\}$

$R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (1,3), (1,4), (2,4)\}$



	1	2	3	4
Indegree	1	2	2	3
Outdegree	4	2	1	1

5. Consider $A = \{1, 2, 3, 4, 6\}$ and R be relation defined by a is related to b iff a is multiple of b. Represent R as a matrix and draw its diagraph.

Solution:

$A = \{1, 2, 3, 4, 6\}$

$R = \{(1,1), (2,2), (3,3), (4,4), (6,6), (4,2), (6,2), (6,3), (2,1), (3,1), (4,1), (6,1)\}$

$$M_R = \begin{matrix} \\ 1 \\ 2 \\ 3 \\ 4 \\ 6 \end{matrix} \begin{matrix} 1 & 2 & 3 & 4 & 6 \end{matrix} \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

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Equivalence relation:

A relation R on set A is said to be an equivalence relation on A if R is reflexive, is transitive on A and R is symmetric.

Ex: $A = \{1, 2, 3, 4\}$ and $R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,3), (3,3), (4,4)\}$

Verify R is equivalence.

→ $(1,1), (2,2), (3,3), (4,4) \in R$

$(a,a) \in R \quad \forall a \in A$.

∴ R is reflexive relation.

$(1,2) \in R$ & $(2,1) \in R$, $(3,4) \in R$ & $(4,3) \in R$.

$(a,b) \in R$ & $(b,a) \in R \quad a,b \in A$

∴ R is symmetric relation.

$(1,2), (2,1), (1,1) \in R$ & $(2,1), (1,2), (2,2) \in R$ & $(3,4), (4,3), (3,3) \in R$.

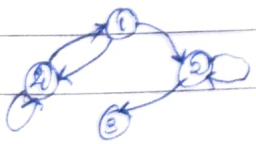
$(a,b) \in R$, $(b,c) \in R$ then $(a,c) \in R \quad a,b,c \in A$

∴ R is transitive on A.

∴ R is an equivalence relation.

b. Find the relation represented by digraph given below also write down its matrix.

(i)

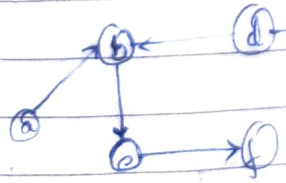


$R = \{(1,2), (1,4), (2,2), (2,3), (4,1), (4,4)\}$

$M_R =$

	1	2	3	4
1	0	1	0	1
2	0	1	1	0
3	0	0	0	0
4	1	0	0	1

(ii)



$R = \{(a,b), (b,c), (d,b), (d,c), (e,f)\}$

$M_R =$

	a	b	c	d	e	f
a	0	1	0	0	0	0
b	0	0	0	0	1	0
c	0	0	0	0	0	0
d	0	1	1	0	0	0
e	0	0	0	0	0	1
f	0	0	0	0	0	0

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Equivalence classes:

Let R be an equivalence relation on set A and $a \in A$ then the set of all those elements $x(a)$ which are related to a by R is called equivalence class.

$R(a)$ or $\bar{a}$ or $[a] = \{x \in A \mid (a,a) \in R\}$

Ex: ① Let $A = \{1,2,3\}$

$R = \{(1,1),(1,3),(2,2),(3,1),(3,3)\}$

$[1] = \{1,3\}$ $[2] = \{2\}$ $[3] = \{1,3\}$

② Let $A = \{1,2,3,4,5,6\}$

$R = \{(1,1),(1,3),(1,5),(3,1),(3,3),(3,5),(5,1),(5,3),(5,5),(2,2),(6,2),(6,6),(4,4)\}$

$[1] = \{1,3,5\}$ $[2] = \{2,6\}$ $[3] = \{1,3,5\}$ $[4] = \{4\}$

$[5] = \{1,3,5\}$ $[6] = \{2,6\}$

③ Let $A = \{1,2,3,4,5\}$

$R = \{(1,1),(2,2),(3,3),(4,4),(5,5),(1,2),(2,1),(4,5),(5,4)\}$

$[1] = \{1,2\}$ $[2] = \{1,2\}$ $[3] = \{3\}$ $[4] = \{4,5\}$ $[5] = \{4,5\}$

Partition of a set:

Let A be an non-empty set suppose there exist a non-empty subset $A_1, A_2, A_3, \ldots, A_k$ such that the following two condition hold

① A is the union of $A_1, A_2, \ldots A_k$ i.e $A = A_1 \cup A_2 \cup A_3 \cup \ldots \cup A_k$

② Any two of the subsets $A_1, A_2, \ldots, A_k$ are disjoint i.e $A_i \cap A_j = \phi$ for $i \neq j$

then the set $P = \{A_1, A_2, A_3, \ldots, A_k\}$ is called a partition of A also $A_1, A_2, A_3, \ldots, A_k$ are called the blocks are cells of the partition.

1. For the equivalence relation $R = \{(1,1), (1,2), (2,1), (2,2), (3,4), (4,3), (3,3), (4,4)\}$ for the set $A = \{1,2,3,4\}$ determine the equivalence classes.

$\Rightarrow A = \{1,2,3,4\}$

$[1] = \{1,2\}$ $[2] = \{1,2\}$ $[3] = \{3,4\}$ $[4] = \{3,4\}$

The equivalence relation of the equivalence class.

$[1]$ and $[3]$ are distinct.

$P = [[1] \cup [3]] = [\{1,2\} \cup \{3,4\}]$

$A = [[1] \cup [3]] = [1,2,3,4]$

2. For $A = \{1,2,3,4,5\}$ the equivalence relation $R = \{(1,1), (2,2), (2,3), (3,2), (3,3), (4,4), (4,5), (5,4), (5,5)\}$ Find the partition of A induced by R.

Solution: $R = \{(1,1), (2,2), (2,3), (3,2), (3,3), (4,4), (4,5), (5,4), (5,5)\}$

$A = \{1,2,3,4,5\}$

$[1] = \{1\}$ $[2] = \{2,3\}$ $[3] = \{2,3\}$ $[4] = \{4,5\}$ $[5] = \{4,5\}$

The equivalence relation of the equivalence class.

The distinct classes here are $[1]$, $[2]$ and $[4]$

Let $P = [[1] \cup [2] \cup [4]] = [\{1\} \cup \{2,3\} \cup \{4,5\}]$

$A[[1] \cup [2] \cup [4]] = [1,2,3,4,5]$

3. $A = \{1,2,3,4,5,6,7\}$.

$A = \{1,2\} \cup \{3\} \cup \{4,5,7\} \cup \{6\}$

Since we've $\{1,2\}$ in block 1

1R1, 1R2, 2R2 2R1.

For block 2

3R3

For block 3 we've $\{4,5,7\}$:

4R4, 4R5, 4R7, 5R4, 5R5, 5R7, 7R4, 7R5, 7R7

For block 4.

6R6

$\therefore R = \{(1,1), (1,2), (2,1), (2,2), (3,3), (4,4), (4,5), (4,7), (5,5), (5,7), (7,4), (6,6), (7,4), (7,5), (7,7)\}$

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4. $A = \{a, b, c, d, e\}$ consider the partition $P = \{(a,b), (c,d), (e)\}$ find the equivalence relation using this partition.

Solution: $A = \{a, b, c, d, e\}$

$P = \{(a,b), (c,d), (e)\}$

$R = \{(a,a), (a,b), (b,a), (b,b), (c,c), (c,d), (d,c), (d,d), (e,e)\}$

Partial orders:

A relation R on set A is said to be a partial ordering relation or partial order on A if R is reflexive, antisymmetric and transitive on A.

The set A with the partial order R defined on it is called partially ordered set or an ordered set or a poset. And it is denoted by (A, R).

Diagraph of partial order is called poset diagraph or hasse diagram.

1. Let $A = \{1, 2, 3, 4\}$, $R = \{(1,1), (1,2), (2,2), (2,4), (1,3), (3,3), (3,4), (4,4)\}$. Verify that R is a partial order on A. Also write the hasse diagram.

Solution:

$(1,1), (2,2), (3,3), (4,4) \in R$

$(a,a) \in R \quad \forall a \in A$

$\therefore$ R is reflexive.

$(a,b) \in R$ but $(b,a) \notin R \quad \forall a, b \in A$.

$\therefore$ R is anti-symmetric

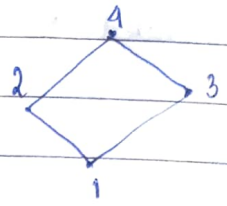
$(1,1)(1,2)(1,3), (2,2)(2,4)(2,4), (3,3)(3,4), (3,4), (1,1)(1,4), (1,4)$

$(a,b) \in R$ and $(b,c) \in R \quad (a,c) \in R \quad \forall a, b, c \in A$

$\therefore$ R is transitive on A.

$\therefore$ R is partial order on A.

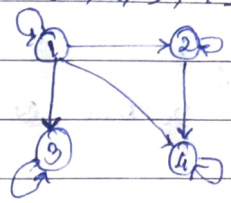
Hesse diagram: – (Do not include reflexive & transitive)



2. A partial order R on the set A is represented by the following diagraph, draw the hesse diagram for R.

Solution:

$A = \{1, 2, 3, 4\}$



$R = \{(1,1), (1,2), (1,3), (1,4), (2,2), (2,4), (3,3), (4,4)\}$

$(1,1), (2,2), (3,3), (4,4) \in R$

$(a,a) \in R \quad \forall \; a \in A$

$\therefore$ R is reflexive

$(a,b) \in R$ but $(b,a) \notin R \quad a,b \in A$

$\therefore$ R is anti-symmetric

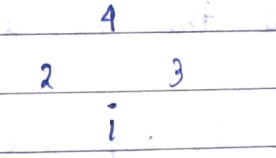
$(1,1)(1,2)(1,2)(1,1)(1,3)(1,3)(1,4)(1,4)(1,4)(2,2)(2,4)(2,4)$

$(a,b) \in R$ & $(b,c) \in R$ then $(a,c) \in R \quad a,b,c \in A$

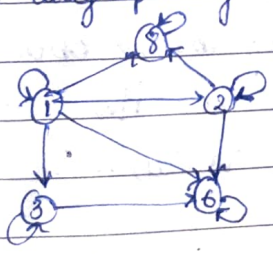
$\therefore$ R is transitive

Hence R is partial order relation.

Hesse diagram:



3. The diagraph of the relation of set $A = \{1, 2, 3, 6, 8\}$



Total order:

A relation R on set A is called a total order if x is related to y (or) y is related to x $\forall x, y \in A$ ~~is called as total order~~.

Topological sorting:

Consider a partial order R on a set A which is not a total order. To find total order T on A such that $R \subset T$, the technique of constructing such a T for a finite set A is known as topological sorting.

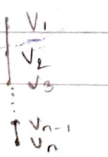
Steps involved in topological sorting:

1. Consider the hasse diagram for the partial order and denote it by H_1
2. Select a vertex in H_1 from which no edge start and denote it by V_1.
3. Delete from H_1 the vertex V_1 as well as all edges of H_1 that terminate at V_1, denote the resulting diagram by H_2.

Repeat the process step 2 & 3 with H_1 replaced by H_2 the vertex considered in step 2 of this repetition is denoted by V_2 and the diagram obtained at the end is denoted by H_3.

Repeat the process until the final diagram is obtained. Suppose the $|H| = n$ the final diagram H_n contains only one vertex V_n with the vertices $V_1, V_2, \ldots, V_{n-1}$ is removed one by one in order.

A relation T on A is defined by $(V_i, V_j) \forall i \geq j$, $j = 1, 2, \ldots$ then T is required total order. The hasse diagram of T is

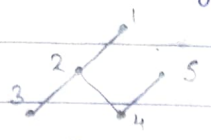


(or) $V_n < V_{n-1} \ldots V_3 < V_2 < V_1$

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1. Consider $A = \{1, 2, 3, 4, 5\}$ and the partial order R on A whose hasse diagram is given below.

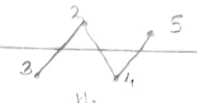
(1)



H_1

$V_1 = 1$

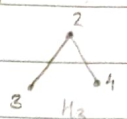
(2)



H_2

$V_2 = 5$ or 2

(3)



H_3

$V_3 = 2$ or 5

(4)



H_4

$V_5 = 4$ or 3

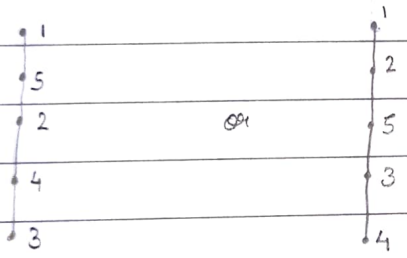
(5)



H_5

$V_5 = 3$ or 4

$V_1 = 1$, $V_2 = 5$ or 2, $V_3 = 2$ or 5, $V_4 = 4$ or 3, $V_5 = 3$ or 4



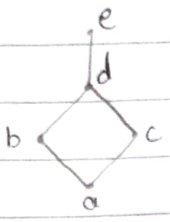
Total order hasse diagram

$R = \{(3,3), (3,4), (3,2), (3,5), (3,1), (4,4), (4,2), (4,5), (4,1), (2,2), (2,5), (2,1), (5,5), (5,1), (1,1)\}$

By comparing H and T we observe that all the ordered pairs belonging to R are in T also T contains the ordered pairs that are not in R $(3,4), (3,5), (2,5), (5,1)$.

$\therefore$ T is the total order on A and $R \subset T$.

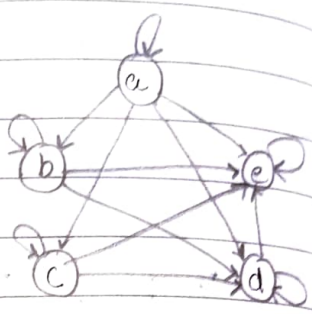
2. Consider $A = \{a, b, c, d, e\}$ (a) Determine the relation matrix and (b) construct the diagram for R (c) Topologically sort the process A, R.



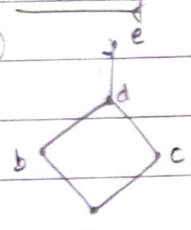
$R = \{(a,a), (a,b), (a,c), (a,d), (a,e), (b,b), (b,d), (b,e), (c,c), (c,d), (c,e), (d,d), (d,e), (e,e)\}$

M_R :

	a	b	c	d	e
a	1	1	1	1	1
b	0	1	0	1	1
c	0	0	1	1	1
d	0	0	0	1	1
e	0	0	0	0	1

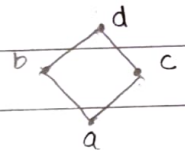


Sorting:

① 

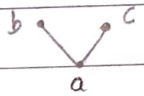
H_1

$V_1 = e$

② 

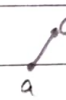
H_2

$V_2 = d$

③ 

H_3

$V_3 = b$ or c

④ 

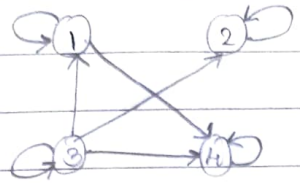
H_4

$V_4 = c$ or b

⑤

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3. $A = \{1, 2, 3, 4\}$ (a) Verify that A, R is a poset and draw its hasse diagram (b) Topologically sort A, R (c) How many more directed edges are needed in the diagraph of R to extend R to an total order.



$R = \{(1,1), (1,4), (2,2), (3,1), (3,2), (3,3), (3,4), (4,4)\}$.

$(a,a) \in R \quad \forall a \in A$.

R is reflexive.

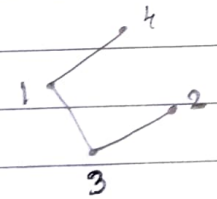
$(a,b) \in R \quad (b,c) \in R$ iff $a = b \quad \forall a, b \in A$.

R is anti-symmetric

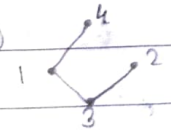
$(a,b) \in R \quad (b,c) \in R$ then $(a,c) \in R \quad \forall a, b, c \in A$.

R is transitive.

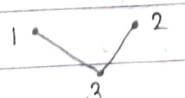
∴ R is partial order on A.



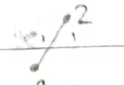
Hasse diagram

① 

$V_1 = 4$

② 

$V_2 = 1$ or 2

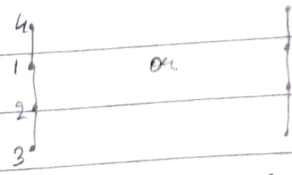
③ 

$V_3 = 2$ or 1

④ 

$V_4 = 3$

$V_1 = 4 \quad V_2 = 1$ or $2 \quad V_3 = 2$ or $1 \quad V_4 = 3$



$T = \{(3,3), (3,2), (3,1), (3,4), (2,2), (2,1), (2,4), (1,1), (1,4), (4,4)\}$

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By comparing R and T we observe that all the ordered pairs belong to R are in T als T contains the ordered pairs that are not in R (2,1), (2,4)
∴ T is total order on A and R ⊂ T.

Operation on relations:

1. Unions (∪)
2. Intersection (∩)
3. Complement of relation ($\overline{R}$)
4. Converse of Relation. (R^c)

Unions of relation:

If A and B are a relation on set A then the union of A and B is denoted as A∪B, is defined as if a, b∈A the (a, b) ∈ A or (a, b) ∈ B.

Ex: A = {1, 2, 3, 4} B = {2, 4, 6, 8}
A∪B = {1, 2, 3, 4, 6, 8}

Intersection:

If A and B are a relations on set A then the intersection the relation A and B is denoted as A∩B, is defined as (a,b)∈A (a,b) ∈ B if (a,b) ∈ A∩B

Ex: A = {1, 2, 3, 4} B = {2, 4, 6, 8}
A∩B = {2, 4}

Complement:

If R is a relation on a set A then the complement of R is denoted as $\bar{R}$ and it is defined as $(a,b) \notin R$.

Ex: $A = \{1, 2, 3, 4\}$

$R = \{(1,2), (2,3)\}$

$A \times A = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$

$\bar{R} = \{(1,1), (1,3), (1,4), (2,1), (2,2), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$

Converse:

If R is a relation on set A then the converse of R is denoted as R^c and it is defined as if $(a,b) \in R$ then $(b,a) \in R^c$.

Ex: $R = \{(1,2), (2,3), (3,4)\}$

$R^c = \{(2,1), (3,2), (4,3)\}$

1. If $A = \{1, 2, 3, 4\}$ and R & S are the relation R is defined $R = \{(1,2), (1,3), (2,4), (4,4)\}$ $S = \{(1,1), (1,2), (1,3), (1,4), (2,3), (2,4)\}$

Solution:

$R \cup S = \{(1,2), (1,3), (1,1), (1,4), (2,3), (2,4), (4,4)\}$

$R \cap S = \{(1,2), (1,3), (2,4)\}$

$A \times A = \{(1,1), (1,2), (1,3), (1,4), (2,1), (2,2), (2,3), (2,4), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$

$\bar{R} = \{(1,1), (1,4), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3), (3,4), (4,1), (4,2), (4,3)\}$

$R^c = \{(2,1), (3,1), (4,2), (4,4)\}$

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2. Consider $A = \{a, b, c\}$ and $B = \{1, 2, 3\}$ and the relations
$R = \{(a,1), (b,1), (c,2), (c,3)\}$
$S = \{(a,1), (a,2), (b,1), (b,2)\}$ from A to B.
Determine $\overline{R}$, $\overline{S}$, $R \cup S$, $R \cap S$, R^c & S^c.

Solution:

$A = \{a, b, c\}$

~~$A \times A = \{(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,a), (c,b), (c,c)\}$~~

$A \times B = \{(a,1), (a,2), (a,3), (b,1), (b,2), (b,3), (c,1), (c,2), (c,3)\}$

$\overline{R} = \{(a,2), (a,3), (b,2), (b,3), (c,1)\}$

$\overline{S} = \{(a,2), (a,3), (b,3), (c,1), (c,2), (c,3)\}$

$R \cap S = \{(a,1), (b,1)\}$

$R \cup S = \{(a,1), (a,2), (b,1), (b,2), (c,2), (c,3)\}$

$R^c = \{(1,a), (1,b), (2,c), (3,c)\}$

$S^c = \{(1,a), (2,a), (1,b), (2,b)\}$

3. Let $A = \{1, 2, 3\}$ and $B = \{1, 2, 3, 4\}$ and the relation R from A to B is represented by matrix.

$$M_R = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix} \qquad M_S = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Determine the relations $\overline{R}$, $R \cup S$, $R \cap S$ and S^c and their matrix representation.

Solution:

$R = \{(1,1), (1,3), (2,4), (3,1), (3,2), (3,3)\}$

$S = \{(1,1), (1,2), (1,3), (1,4), (2,4), (3,2), (3,4)\}$

$\overline{R} = \{(1,2), (1,4), (2,1), (2,2), (2,3), (3,4)\}$

$R \cup S = \{(1,1), (1,2), (1,3), (1,4), (2,4), (3,1), (3,2), (3,3), (3,4)\}$

$R \cap S = \{(1,1), (1,3), (2,4), (3,2)\}$

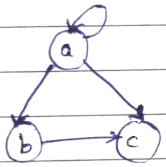
$S^c = \{(1,1), (2,1), (3,1), (4,1), (4,2), (2,3), (4,3)\}$

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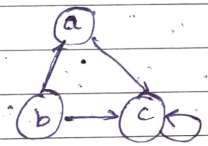
$M_{\bar{R}} = \begin{bmatrix} 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$ $M_{R \cup S} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$

$M_{R \cap S} = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 \end{bmatrix}$ $M_{S^c} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 \end{bmatrix}$

The diagraph of 2 relations R and S on a set A.
A = {a, b, c} Determine $\bar{R}$, R∪S, R∩S and R^c



R



S

Solution:

A = {a, b, c}

A×A = {(a,a), (a,b), (a,c), (b,a), (b,b), (b,c), (c,a), (c,b), (c,c)}

R = {(a,a), (a,b), (a,c), (b,c)}

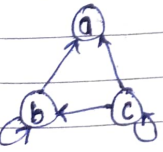
S = {(a,c), (b,a), (b,c), (c,c)}

$\bar{R}$ = {(b,a), (b,b), (c,a), (c,b), (c,c)}

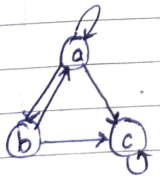
R∪S = {(a,a), (a,b), (a,c), (b,a), (b,c), (c,c)}

R∩S = {(a,c), (b,c)}

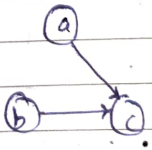
R^c = {(a,a), (b,a), (c,a), (c,b)}



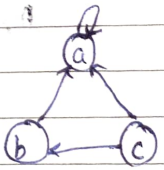
$\bar{R}$



R∪S



R∩S



R^c

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Composition of relation (∘):

If R and S are the two relation on a non-empty set then the composition relation of R and S is denoted $R \circ S$ and it is defined as if $(a,b) \in R$ and $(b,c) \in S$ then $(a,c) \in R \circ S$

Let $A = \{1,2,3,4\}$ and $R = \{(1,1),(1,2),(2,3),(3,4)\}$ and $S = \{(3,1),(4,1),(2,4),(1,4)\}$ be a relation on A.
Determine $R \circ S$, $S \circ R$, R^2 and S^2.

Solution:

$R \circ S = \{(1,4),(2,1),(3,4)\}$
$S \circ R = \{(3,1),(3,2)\}$
$R^2 = \{(1,2),(1,3),(2,4),(1,1)\}$
$S^2 = \{(3,4),(4,4),(1,4),(2,4)\}$

If $A = \{1,2,3,4\}$ and R and S are relations on A defined by $R = \{(1,2),(1,3),(2,4),(3,4)\}$ and $S = \{(1,1),(1,2),(1,3),(1,4),(2,3),(2,4)\}$.
Find $R \circ S$, $S \circ R$, R^2, S^2. Represent them in matrix.

Solution:

$R \circ S = \{(1,3),(1,4)\}$
$S \circ R = \{(1,2),(1,3),(1,4),(2,4)\}$
$R^2 = \{(1,4),(2,4),(4,4)\}$
$S^2 = \{(1,1),(1,2),(1,3),(1,4)\}$

$M_{R \circ S} = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $M_{S \circ R} = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ $M_{R^2} = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

$M_{S^2} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

3. Let $A = \{a, b, c\}$

$$M_R = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix} \qquad M_S = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Determine $R \circ S$, $S \circ R$, $R \circ R$, $S \circ S$

4. Let $R = \{(1,1), (1,2), (2,3), (3,3), (3,4)\}$ $A = \{1, 2, 3, 4\}$.

Extreme elements in poset :

1. Maximal element.
2. Minimal element.
3. Least element
4. Greatest element.
5. Upper bound
6. Lower bound
7. Greatest lower bound [GLB]
8. Least upper bound [LUB]

Poset $P(A, R)$ the extreme elements in a poset is defined as follows.

1. Maximal element :

An element $a \in A$ is called the maximal element of A if there exist no element $x \neq a$ in A such that a is related to x. (R)
in an hasse diagram of R a maximal element of A means that no edge starts from a.

2. Minimal element :

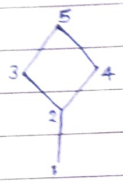
An element $a \in A$ is called a minimal element of A if there exist no element $x \neq a$ in A such that x is related a (R)
in a hasse diagrams of R the minimal element of A means that

no edge terminates from a.

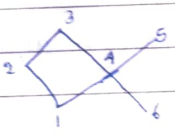
3. Greatest element:
An element $a \in A$ is called a greatest element of A if x related a for all $x \in A$.

4. Least element:
An element $a \in A$ is called a least element of A if a is related x $\forall$ $x \in A$.

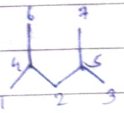
1. Find maximal, minimal, greatest and least element of a given hasse diagram.

(i) 

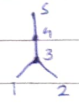
Maximal element - 5
Minimal element - 1
Least element - 1
Greatest element - 5

(ii) 

Maximal - 3 & 5
Minimal - 1 & 6
Least - -
Greatest - -

(iii) 

Max - 6 & 7
Min - 1 & 2 & 3

(iv) 

Max - 5
Min - 1 & 2

Gr - 5

(v) 

Max - 6
Min - 1

- Gr - 6
L - 1

Upper bound element :

An element $a \in A$ is called upper bound of the subset $B(a)$ if x is related a $\forall x \in A$.

Lower bound element :

An element $a \in A$ is called lower bound of the subset $B(a)$ if a is related x $\forall x \in B$

Least upper bound element :

An element $a \in A$ is called least upper bound of a subset $B(a)$

(i) if a is an upperbound of B.

(ii) if d is an upperbound of B, then a is related to d.

Least upper bound element is called as suppreme of :- Sup

Greatest lower bound :

An element $a \in A$ is called greatest lower bound of a subset $B(a)$

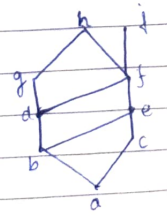
(i) a is lower bound of B

(ii) if d is a lower bound of B then d is related a.

Greatest lower bound is also called as infimem

① $A = \{a, b, c, d, e, f, g, h, i, j\}$

$B = \{a, b, c\}$ $\quad B \to A$



aRa, aRb, aRc, aRd, aRe, aRf, aRg, aRh, aRj

bRa, bRb, bRc, bRd, bRe, bRf, bRg, bRh, bRj

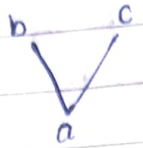
cRa, cRb, cRc, cRd, cRe, cRf, cRg, cRh, cRj

Upperbound = $\{e, f, h, i, j\}$

Least upper bound = $\{e\}$

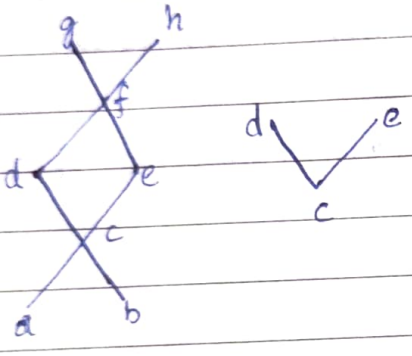
$A \to B$

aRa aRb aRc
bRa bRb bRc
cRa cRb cRc
dRa dRb dRc
eRa eRb eRc
fRa fRb fRc
gRa gRb gRc
hRa hRb hRc
jRa jRb jRc



Lower bound = {a}
Greatest lower bound = {a}

e.



A = {a, b, c, d, e, f, g, h}
B = {c, d, e}

A → B

aRc aRd aRe
bRc bRd bRe
cRc cRd cRe
dRc dRd dRe
eRc eRd eRe
fRc fRd fRe
gRc gRd gRe
hRc hRd hRe

Lower bound = {a,b,c
Greatest lower bound

$B \to A$

cRa cRb cRc cRd cRe cRf cRg cRh
dRa dRb dRc dRd dRe dRf dRg dRh
eRa eRb eRc eRd eRe eRf eRg eRh

Upper bound = {f, g, h}
Lowest upper bound = {f}